

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.21)

Metric

$$ds^2 = -dt \otimes dt + 2y dt \otimes dz + \frac{a^2}{x^2} dx \otimes dx + \frac{a^2}{x^2} dy \otimes dy + 2y dz \otimes dt + (x^2 - 4y^2) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{1}{2} \sqrt{2} (x + 2y) \right) dz$$

$$n = \left(\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{1}{2} \sqrt{2} (x - 2y) \right) dz$$

$$m = \left(\frac{\sqrt{2}a}{2x} \right) dx + \left(\frac{i\sqrt{2}a}{2x} \right) dy$$

$$\bar{m} = \left(\frac{\sqrt{2}a}{2x} \right) dx + \left(-\frac{i\sqrt{2}a}{2x} \right) dy$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}(x + 2y)}{2x} \right) dt + \left(-\frac{\sqrt{2}}{2x} \right) dz$$

$$n = \left(-\frac{\sqrt{2}(x-2y)}{2x} \right) dt + \left(\frac{\sqrt{2}}{2x} \right) dz$$

$$m = \left(\frac{\sqrt{2}x}{2a} \right) dx + \left(\frac{i\sqrt{2}x}{2a} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}x}{2a} \right) dx + \left(-\frac{i\sqrt{2}x}{2a} \right) dy$$

Directional Derivatives

$$D = -\frac{\sqrt{2}(x+2y)}{2x} \partial_t - \frac{\sqrt{2}}{2x} \partial_z$$

$$\Delta = -\frac{\sqrt{2}(x-2y)}{2x} \partial_t + \frac{\sqrt{2}}{2x} \partial_z$$

$$\delta = \frac{\sqrt{2}x}{2a} \partial_x + \frac{i\sqrt{2}x}{2a} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}x}{2a} \partial_x - \frac{i\sqrt{2}x}{2a} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = \frac{(2i+1)\sqrt{2}}{4a}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = -\frac{(2i+1)\sqrt{2}}{4a}$$

$$\alpha = -\frac{(i-1)\sqrt{2}}{4a}$$

$$\beta = \frac{(i-1)\sqrt{2}}{4a}$$

$$\tau = -\frac{\sqrt{2}}{4a}$$

$$\pi = \frac{\sqrt{2}}{4a}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{a^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{1}{a^2}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{1}{a^2}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{a^2}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = -\frac{i+3}{2a^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{i-1}{2a^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{i+3}{2a^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"